BACHELOR THESIS
COGNITIVE INFORMATICS

AUTOMATIC FEEDBACK FOR MOTION EXECUTION ERRORS BASED ON SMARTPHONE VIDEO DATA

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The real risk is making sure that proper form is maintained, and that you lift the amount your body can handle and not what your ego thinks you can handle.

— ARNOLD SCHWARZENEGGER

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1 INTRODUCTION

Incorrect movement execution during strength training is a major cause of preventable injuries [Bon+22]. Proper technique requires continuous feedback, which is typically provided by experienced coaches or professional motion analysis systems [WF23]. However, human coaching is not always available and high-precision motion capture systems are expensive and impractical for everyday use [Wan+26; Che+25]. This raises the question of how automatic, accessible, and reliable movement feedback can be provided using widely available consumer devices such as smartphones.

In particular, this thesis focuses on automatic feedback generation for squat execution based on single-camera smartphone video recordings. The squat is a fundamental compound movement that involves coordinated hip, knee, and ankle movements and is often performed incorrectly [SP24]. Detecting execution errors from monocular video data is challenging due to depth ambiguity, occlusions, varying lighting conditions, and noisy joint detections [Ji+20; Guo+25]. Furthermore, many existing machine learning approaches act as black-box systems, limiting the interpretability of the resulting feedback [Kra+25].

Previous work has addressed human motion analysis using multi-camera motion capture systems such as those developed by VICON, which provide highly accurate 3D pose estimation. Although these systems are precise, they require controlled environments and specialized hardware, making them unsuitable for low-cost and mobile applications [AWW25; Mer+17]. More recently, markerless human pose estimation frameworks such as OpenPose and MediaPipe have enabled real-time keypoint detection from single RGB cameras [Cao+17]. However, these approaches focus primarily on pose detection rather than robust biomechanical analysis and interpretable feedback generation under real-world recording conditions. In particular, instability in joint localization, visibility fluctuations, and temporal noise often lead to unreliable angle estimation, which directly affects the quality of movement feedback [Pav+19; Sar+16].

To address these limitations, this thesis proposes a markerless motion analysis pipeline designed specifically for stability-aware squat assessment using single-camera smartphone video data. The approach integrates visibility-based keypoint gating, adaptive temporal filtering, and biomechanically motivated joint angle computation to produce robust and interpretable movement parameters. Instead of relying on opaque machine learning classification outputs, the system derives explicit biomechanical metrics that form the basis for rule-based feedback generation.

The main contributions of this thesis are:

1. A complete single-camera, markerless motion analysis pipeline tailored to smartphone recordings

2. A stability-aware joint angle estimation strategy based on visibility gating and adaptive filtering
3. A biomechanically interpretable parameter extraction framework for squat assessment
4. A rule-based feedback mechanism that translates quantitative movement metrics into actionable correction suggestions
5. An evaluation of the proposed system under realistic recording conditions

The remainder of this thesis is structured as follows. Chapter 2 reviews related work on biomechanics, pose estimation, and video-based motion analysis. Chapter 3 presents the proposed method. Chapter 4 describes the experimental evaluation. Finally, Chapter 5 summarizes the main findings and outlines directions for future work.

2 BACKGROUND AND RELATED WORK

2.1 BIOMECHANICS OF THE SQUAT

The squat is one of the most commonly used exercises in strength training and rehabilitation due to its ability to simultaneously engage multiple lower extremity muscle groups, as shown in Figure 2.1 [Esc01]. During squat execution, primary joint motions occur in the sagittal plane and involve coordinated flexion and extension of the hip, knee, and ankle joints [Sch10]. The biomechanical characteristics of the squat have been extensively studied in the literature of sports science.

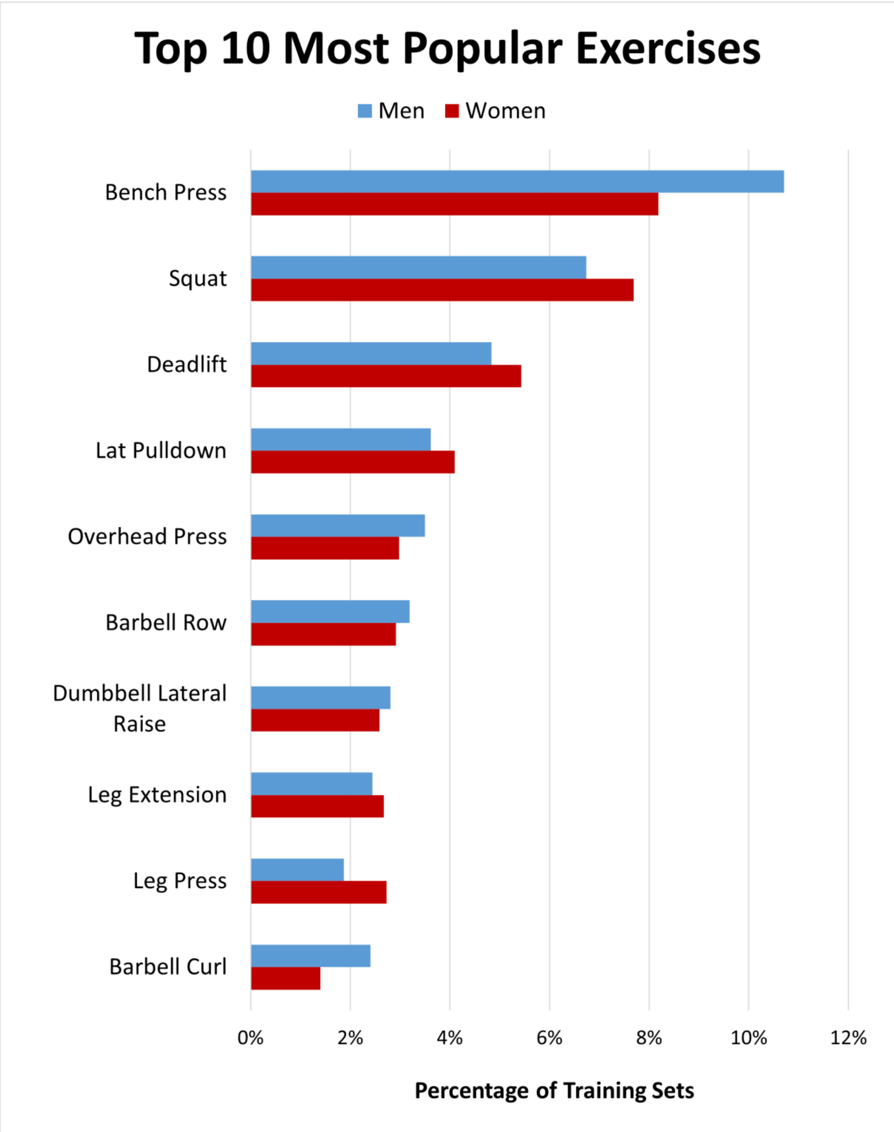

Figure 2.1: Most Popular Exercises [Ric]

[Esc01] analyzed the biomechanics of the knee during dynamic squatting and showed that joint loading and muscle activation patterns are strongly dependent on the depth and technique of the squat. Similarly, [Sch10] summarized the kinematic and kinetic characteristics of squat movements and emphasized the importance of proper technique to reduce the risk of injury and improve training effectiveness. Variations in stance width, trunk inclination, and joint angles can significantly alter joint loading patterns and muscle recruitment during the exercise.

[Lor+18] further demonstrated that the squat technique varies substantially between individuals and is influenced by factors such as stance width and foot placement. Their findings highlight that the joint angles of the hip, knee, and trunk provide meaningful indicators to evaluate squat execution. Consequently, joint angle analysis has become a common method for assessing movement quality and detecting potential technical deficiencies in squat performance.

Because the majority of squat motion occurs in the sagittal plane, side-view observations allow meaningful approximation of hip, knee, and ankle joint kinematics. This characteristic makes squats particularly suitable for analysis using two-dimensional motion capture approaches.

2.2 HUMAN POSE ESTIMATION

Human pose estimation aims to infer the positions of human body joints from image or video data. In most modern approaches, the human body is represented as a collection of keypoints that correspond to anatomical landmarks such as the shoulders, hips, knees, and ankles.

Earlier pose estimation techniques relied mainly on handcrafted visual features and probabilistic graphical models [CY14]. However, with the rise of deep learning, the accuracy and robustness of these systems have improved significantly. One of the most influential methods in this area is OpenPose, introduced by [Cao+17]. This framework enables real-time multi-person pose estimation by using part affinity fields to detect individual body joints and determine how they belong together.

More recent research has focused on lightweight models that can run efficiently on mobile or embedded devices. A notable example is the MediaPipe Pose (MPP) framework proposed by [Baz+20]. MPP provides real-time full-body pose estimation while maintaining high computational efficiency, making it well suited for applications on consumer hardware. The system detects 33 body landmarks and has been used in a variety of contexts, including sports analysis, gesture recognition, and human–computer interaction.

Most pose estimation frameworks output normalized landmark coordinates along with confidence or visibility scores that indicate how reliable each detected landmark is. These keypoints can then be used to compute higher-level biomechanical measures such as joint angles or movement trajectories [Ji+20].

2.3 MARKER-BASED VS MARKERLESS MOTION CAPTURE

Traditional biomechanical motion analysis is commonly performed using marker-based motion capture systems such as Vicon. These systems rely on reflective markers attached to the body and multiple high-speed cameras to reconstruct three-dimensional joint positions. While marker-based systems provide high accuracy, they require specialized laboratory environments, extensive calibration procedures, and significant financial investment [Mer+17].

Recent advances in computer vision have enabled markerless motion capture methods that estimate body kinematics directly from video recordings. Markerless systems offer several advantages, including reduced setup time, lower costs, and increased accessibility for non-laboratory environments. Deep learning–based frameworks such as DeepLabCut [Mat+18] have demonstrated that markerless pose estimation can achieve high accuracy in many applications.

[Col+18] provide a comprehensive review of vision-based motion analysis systems and discuss the growing integration of deep learning techniques into markerless motion capture pipelines. Furthermore, [Kan+21] showed that markerless motion capture systems can produce kinematic measurements comparable to traditional marker-based systems for certain movement tasks.

These developments have opened up new possibilities for movement analysis using consumer-grade cameras and mobile devices.

2.4 VIDEO-BASED EXERCISE ANALYSIS

Computer vision techniques have increasingly been applied to analyze physical exercises using video recordings. Many existing systems rely on pose estimation to extract joint trajectories that can subsequently be used for activity recognition or movement evaluation [Col+18].

Several studies have proposed systems that classify exercises such as squats, push-ups, or lunges based on pose-derived features. For example, [CY20] demonstrated that pose estimation combined with machine learning techniques can successfully recognize different exercise types from video data. Such systems typically focus on activity classification rather than detailed evaluation of movement quality.

More recent research has begun to explore automatic feedback generation based on biomechanical features extracted from pose estimation data. These systems aim to identify technical errors in exercise execution and provide corrective feedback to the user. However, many existing approaches rely on complex machine learning models that may lack interpretability or require large labeled datasets for training [Kra+25]. Compared to machine-learning-based approaches, rule-based systems offer improved interpretability but may suffer from limited flexibility [Rud19; RSG16].

2.5 TEMPORAL FILTERING IN MOTION ANALYSIS

Pose estimation outputs are often affected by noise, jitter, and occasional tracking failures, which can lead to unstable joint angle trajectories [Cao+17]. Temporal filtering techniques are therefore commonly applied to smooth pose signals and improve measurement stability.

Traditional filtering approaches in biomechanics include low-pass filters such as Butterworth filters or moving average filters [Win09]. However, these methods may introduce delays or oversmoothing during rapid movements.

The One Euro filter, introduced by [CRV12], is an adaptive low-pass filter designed for real-time interactive systems. The filter dynamically adjusts its cutoff frequency based on the rate of signal change, allowing it to reduce noise during slow movements while maintaining responsiveness during faster movements. This property makes the One Euro filter particularly suitable for smoothing pose estimation signals in video-based motion analysis.

2.6 SUMMARY OF RESEARCH GAP

Although recent advances in pose estimation and markerless motion capture have enabled new possibilities for video-based movement analysis, several challenges remain. Many existing systems require specialized laboratory setups [Col+18] or rely on complex machine learning models that do not adequately interpret the underlying biomechanical causes [Kra+25]. Furthermore, relatively few approaches focus specifically on generating interpretable feedback using accessible consumer devices.

This work addresses these limitations by proposing a markerless squat analysis system that operates on smartphone video recordings and provides rule-based feedback derived from biomechanical joint-angle relationships. By combining pose estimation, signal processing, and interpretable decision rules, the proposed method aims to provide an accessible and reproducible approach for automatic movement assessment outside laboratory environments.

3 METHODS

3.1 SYSTEM OVERVIEW

This work proposes a markerless video-based system for the automatic assessment of squat execution using side-view smartphone recordings. The objective of the system is to extract kinematic joint parameters from 2D pose estimation outputs and transform them into interpretable technique feedback related to movement quality.

The proposed method follows a structured multi-stage processing pipeline that transforms raw RGB video data into biomechanical features and rule-based feedback outputs. The overall architecture of the system is illustrated in Figure 3.1.

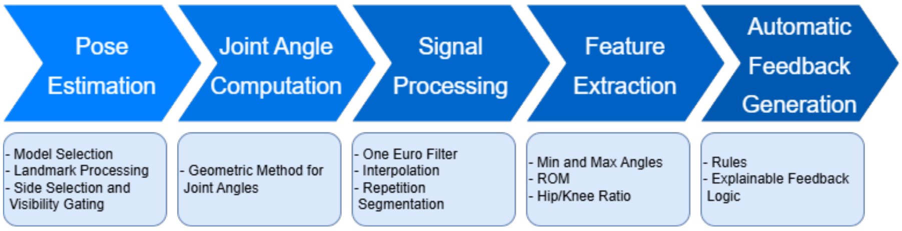

Figure 3.1: Processing Pipeline

Conceptual Data Flow:
The system processes video frame-by-frame. For each frame, body landmarks are extracted using a pre-trained 2D pose estimation model. From these landmarks, sagittal-plane joint angles are computed using geometric relationships. The resulting angle time series are temporally filtered to reduce jitter and stabilize noise measurement. Subsequently, the continuous motion sequence is segmented into individual squat repetitions based on the knee angle trajectory. For each detected repetition, biomechanical features are computed and evaluated against predefined decision criteria. These criteria translate kinematic relationships into interpretable movement feedback.

3.2 DATA ACQUISITION

The data acquisition process can be divided into two main parts. The first part consists of the video recordings. During this stage, several considerations and preparation steps were taken to ensure that the recorded data is suitable for pose estimation and provides reliable input for the later analysis.

At the same time, the recorded videos are also used to create precisely hand-annotated reference data. These annotations serve as ground truth and allow an evaluation of how accurately the pose estimation system detects the relevant body landmarks.

3.2.1 Video Recordings

The video recordings used in this study were captured using an Apple iPad positioned to obtain a side-view perspective of the participants performing squats. The camera was placed approximately perpendicular to the movement direction in order to approximate a sagittal plane view (approximately 90°). Figure 3.2 illustrates five frames from a recording of a participant performing a squat.

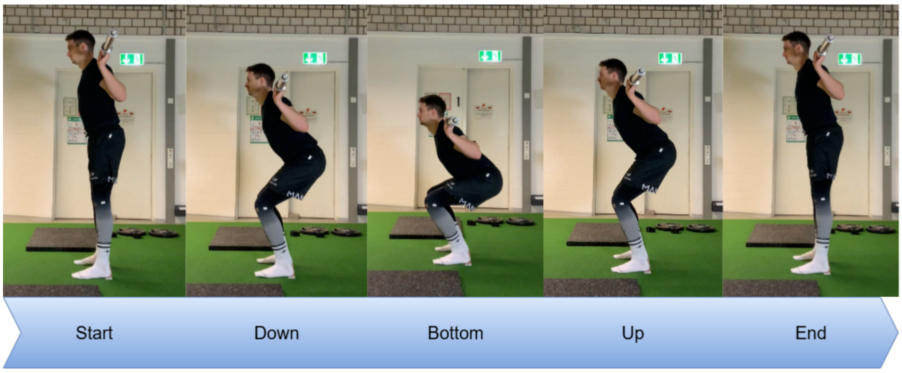

Figure 3.2: Video Recordings of a Participant

The side-view perspective was chosen because it provides a reliable camera angle for detecting the key landmarks required for the analysis, including the shoulder, hip, knee, ankle, and foot index. This perspective allows the main joint movements of the squat, which primarily occur in the sagittal plane, to be observed and analyzed. Although the recordings in this study were captured using an iPad, the proposed approach is not limited to this device and can be applied using other consumer devices such as smartphones from different manufacturers.

The recording device was placed upright at a height of approximately 1 meter above the ground and positioned at a distance that ensured the participant's entire body remained visible during the exercise. All recordings were performed using a standard frame rate of 30 frames per second.

The recordings were conducted in an indoor environment with sufficient lighting to allow reliable landmark detection by the pose estimation model. No reflective markers or additional tracking equipment were used, as the system relies entirely on markerless pose estimation from video data. Participants were asked to wear tight-fitting sports clothing to improve the quality of the data while performing squats without external constraints.

To ensure realistic training conditions, participants performed back squats with a moderate load between 20 kg and 40 kg. The additional load was intended to place the body under realistic mechanical tension, which results in movement patterns closer to those observed during typical strength training. Each participant performed multiple squat repetitions while maintaining a natural and self-selected movement tempo.

3.2.2 Reference Data

The reference data was collected using a commonly used software tool in sports science called Kinovea. Kinovea is a video analysis application designed specifically for motion analysis in sports.

The software provides several useful features, including the ability to capture, slow down, compare, annotate, track, and measure movement in video recordings. In this thesis, the annotation and tracking functions were used to manually label relevant body positions and obtain joint angles that closely represent the actual movements of the participants.

These manually annotated measurements serve as reference data and allow a comparison between the pose estimation results and the angles observed in the recorded movements.

3.3 POSE ESTIMATION

3.3.1 Model Selection

In this thesis, the MediaPipe Pose (MPP) framework was used for human pose estimation. MPP is a markerless pose estimation framework that is capable of estimating 3D body landmarks from image or video data. The model detects multiple keypoints of the human body and tracks them across frames, which allows the reconstruction of body posture during movement.

Although MPP provides three-dimensional landmark coordinates, this work only uses the two-dimensional image coordinates of the detected landmarks. This decision was made because the recordings were captured from a fixed side-view perspective, which already approximates the sagittal plane of the movement. Since the goal of this work is to extract joint angles of the hip, knee, and ankle for squat analysis, the two-dimensional projection of the landmarks is sufficient for estimating these angles.

An advantage of MPP is its detailed set of detected landmarks. Unlike some other pose estimation frameworks, such as OpenPose, which represent the foot using only a single landmark, MPP provides separate landmarks for the ankle, heel, and foot index. This more detailed landmark representation makes it possible to compute the ankle joint angle, which would not be feasible if the foot were represented by only a single point.

Another advantage of MPP is its ability to perform pose estimation in real time while maintaining relatively high accuracy. This makes it suitable for applications that aim to provide movement feedback during or shortly after exercise execution.

Additionally, the framework is designed to run efficiently on mobile devices and smartphones. While this thesis processes the recorded videos in a Python environment, the mobile compatibility of MPP makes it a promising option for future systems that could provide real-time feedback directly on portable devices.

3.3.2 *Landmark Processing*

MPP is capable of detecting and tracking 33 body landmarks. These landmarks correspond to different anatomical points of the human body, such as joints and extremities.

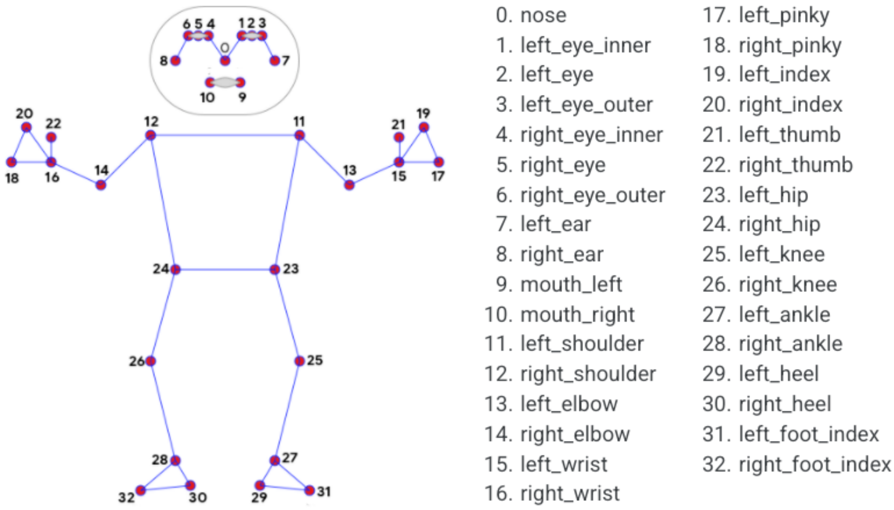

Figure 3.3: Pose Landmarks [Baz+20]

For the purposes of this thesis, only a subset of these landmarks is required for the analysis of squat movements. In particular, the landmarks corresponding to the shoulder, hip, knee, ankle and foot index (tip of the foot) are used to compute joint angles relevant to squat mechanics.

MPP provides landmarks for both the left and right side of the body. The landmark indices used in this work are:

Left body landmarks:

- 11 – Left shoulder
- 23 – Left hip
- 25 – Left knee
- 27 – Left ankle
- 31 – Left foot index

Right body landmarks:

- 12 – Right shoulder
- 24 – Right hip
- 26 – Right knee

- 28 – Right ankle
- 32 – Right foot index

Although the system is capable of processing both sides of the body, the recordings collected for this thesis primarily show the left side of the athlete, due to the chosen camera perspective. Therefore, the left-side landmarks are mainly used for the subsequent analysis.

The extracted landmarks are provided by MPP in normalized image coordinates, where the values of x and y range between 0 and 1 relative to the image dimensions. In order to compute geometric relationships such as joint angles, these normalized coordinates must be converted into pixel coordinates.

This conversion is performed using the following transformation:

$$x_{pixel} = x_{norm} * width$$
$$y_{pixel} = y_{norm} * height$$

where width and height represent the dimensions of the video frame in pixels.

Although angle calculations in a side-view perspective are scale-invariant and would also work with normalized coordinates, the conversion simplifies visualization and ensures compatibility for geometric calculations such as vector operations and joint angle computation, which leads us to the following section.

3.3.3 Side Selection and Visibility Gating

The goal of this step is to ensure consistent landmark tracking throughout the video and to increase the stability of the extracted motion data. Since MPP detects landmarks for both the left and right side of the body, inconsistencies can occur if the visibility of one side changes during the recording. In such cases, the detected landmarks may switch between sides, which would result in unstable joint angle calculations and irregular motion trajectories.

To prevent this, the system first determines which side of the body is more clearly visible in the recording and then consistently uses that side for further analysis. In the recordings used in this thesis, the left side of the body is typically more visible due to the chosen camera perspective. For this reason, the subsequent analysis mainly relies on the landmarks of the left side.

In addition, a visibility gating step is applied to remove unreliable landmark detections. MPP provides a visibility score for each detected landmark that indicates how confident the model is in the detection. If the average visibility of the relevant landmarks falls below a predefined threshold, the corresponding frame is considered unreliable and is excluded from further processing.

This filtering step helps prevent noisy or incorrect landmark detections from affecting the computed joint angles. Frames that do not pass the visibility check can later be handled through interpolation during the signal processing stage. However, if too many frames are discarded, this can also reduce the reliability of the final feedback produced by the system.

3.4 JOINT ANGLE COMPUTATION

To analyze the squat movement, joint angles of the hip, knee, and ankle must be calculated from the detected body landmarks. These joint angles are determined using geometric relationships between three connected landmarks.

For each joint, three landmarks A, B, and C are used, where B represents the joint at which the angle should be calculated. For example, when calculating the knee angle, the landmarks correspond to the hip, knee, and ankle. The angle at point B is determined by computing the angle between the two vectors that connect the joint to its neighboring landmarks.

The first step is to construct the direction vectors from the central landmark B to the surrounding landmarks A and C. These vectors represent the two body segments connected at the joint. Next, the magnitude (or length) of each vector is calculated. The vector magnitudes represent the lengths of the body segments in the image plane. The cosine of the angle between the two vectors can be computed using the dot product formula. This equation describes the relationship between the dot product of the vectors and the angle between them. The angle itself can then be obtained by applying the inverse cosine function. This yields the joint angle in radians.

$$\text{Given three points } A(a_x, a_y),\ B(b_x, b_y),\ C(c_x, c_y)$$

Step 1: Construct vectors originating at B

The first step is to construct the direction vectors from the central landmark B to the surrounding landmarks A and C.

$$\vec{BA} = A - B = \begin{pmatrix} a_x - b_x \\ a_y - b_y \end{pmatrix}, \qquad \vec{BC} = C - B = \begin{pmatrix} c_x - b_x \\ c_y - b_y \end{pmatrix}$$

These vectors represent the two body segments connected at the joint.

Step 2: Compute the vector magnitudes

Next, the magnitude (or length) of each vector is calculated.

$$||\vec{BA}|| = \sqrt{(a_x - b_x)^2 + (a_y - b_y)^2}, \qquad ||\vec{BC}|| = \sqrt{(c_x - b_x)^2 + (c_y - b_y)^2}$$

The vector magnitudes represent the lengths of the body segments in the image plane.

Step 3: Compute the cosine of the angle

The cosine of the angle between the two vectors can be computed using the dot product formula.

$$\cos(\theta) = \frac{\vec{BA} \cdot \vec{BC}}{||\vec{BA}||\,||\vec{BC}||}$$

This equation describes the relationship between the dot product of the vectors and the angle between them.

Step 4: Compute the angle

The angle itself can then be obtained by applying the inverse cosine function.

$$\theta = \arccos\left(\frac{\vec{BA} \cdot \vec{BC}}{||\vec{BA}||\,||\vec{BC}||}\right)$$

This yields the joint angle in radians.

Step 5: Convert to degrees

For easier interpretation and comparison, the angle is converted from radians to degrees.

$$\theta_{\text{deg}} = \theta \cdot \frac{180}{\pi}$$

Using this method, joint angles for the hip, knee, and ankle are calculated for every frame of the video. The specific landmark combinations used in this work are:

- Hip angle: shoulder – hip – knee
- Knee angle: hip – knee – ankle
- Ankle angle: knee – ankle – foot index

These angles form the basis for the movement analysis performed in later sections, where the angle trajectories are further processed to segment squat repetitions and generate feedback about movement execution.

3.5 SIGNAL PROCESSING

Pose estimation results often contain noise and small fluctuations between consecutive frames. These variations can occur for several reasons, such as temporary errors in landmark detection, changes in lighting conditions, or minor inaccuracies in the pose estimation model itself. If the raw landmark positions were used directly, these fluctuations would lead to jittery and unstable joint angle calculations.

To reduce this noise while still preserving the natural dynamics of the movement, an adaptive low-pass filter known as the One Euro Filter was applied. The One Euro Filter, introduced by Casiez et al. (2012), is widely used in interactive applications to smooth noisy signals while still reacting quickly to changes in motion.

In contrast to traditional low-pass filters with a fixed cutoff frequency, the One Euro Filter adjusts the amount of smoothing depending on how quickly the signal changes. Slower movements are smoothed more strongly to remove noise, whereas faster movements are filtered less to avoid introducing noticeable delay.

3.5.1 *One Euro Filter*

Let x_t represent the current signal value at time t. In this work, the signal corresponds to the joint angle measurements extracted from the pose estimation model.

Step 1: Compute the smoothing coefficient alpha

The time difference between two consecutive observations is defined as

$$\Delta t = t - t_{t-1}$$

The smoothing coefficient α is computed based on the cutoff frequency f and the time step Δt:

$$\alpha(f, \Delta t) = \frac{1}{1 + \frac{1}{2\pi f \Delta t}}$$

Step 2: Derivative Estimation

The filter first estimates the rate of change of the signal, which represents how quickly the signal is moving.

$$\dot{x}_t = \frac{x_t - \hat{x}_{t-1}}{\Delta t}$$

This derivative is then smoothed using another low-pass filter:

$$\hat{\dot{x}}_t = \alpha(f_d, \Delta t)\dot{x}_t + \big(1 - \alpha(f_d, \Delta t)\big)\hat{\dot{x}}_{t-1}$$

Step 3: Adaptive cutoff frequency

The smoothed derivative is used to determine an adaptive cutoff frequency for the main filter:

$$f_{c,t} = f_{\min} + \beta|\hat{\dot{x}}_t|$$

This equation allows the filter to automatically adjust its smoothing strength depending on the speed of the signal.

Step 4: Final filtered signal

Finally, the filtered signal value is computed as

$$\hat{x}_t = \alpha(f_{c,t}, \Delta t)x_t + \big(1 - \alpha(f_{c,t}, \Delta t)\big)\hat{x}_{t-1}$$

The resulting signal $\hat{x}_t$ represents the smoothed joint angle value used for further analysis.

In this thesis, the One Euro Filter is applied to the extracted hip, knee, and ankle joint angles. This filtering step reduces frame-to-frame noise while preserving the overall shape and dynamics of the squat movement. As a result, the joint angle trajectories become more stable and suitable for further analysis.

3.5.2 Interpolation

During pose estimation, some frames may contain missing or unreliable joint angle values. This can happen when landmarks are temporarily occluded, when the pose estimation model fails to detect a landmark, or when the visibility threshold discards unreliable measurements. As a result, short gaps can appear in the joint angle time series.

To avoid disruptions in the signal processing pipeline, missing values or NANs (not a number) are filled using linear interpolation. Linear interpolation estimates the missing value between two valid observations by assuming a linear transition between them.

Let x_{t1} and x_{t2} represent two valid signal values surrounding a missing observation at time t, where $t_1 < t < t_2$. The interpolated value $\tilde{x}_t$ can be computed as

$$\tilde{x}_t = x_{t_1} + \frac{t - t_1}{t_2 - t_1} \left(x_{t_2} - x_{t_1} \right)$$

Where:

- x_{t_1} and x_{t_2} are valid signal values surrounding the missing observation
- t_1 and t_2 denote the corresponding timestamps
- t represents the timestamp of the missing value
- $\tilde{x}_t$ is the interpolated signal value

This approach creates a smooth transition between valid measurements and prevents artificial jumps in the signal.

However, interpolation is only applied to short gaps in the data. If too many consecutive frames are missing, the interpolated values may no longer accurately represent the actual movement. Therefore, a maximum interpolation length is defined. If the number of missing frames exceeds this threshold, the gap remains unfilled and the corresponding frames are treated as invalid.

Limiting the interpolation length ensures that the signal reconstruction remains realistic while still allowing small gaps in the data to be corrected.

The interpolated and filtered angle signals are then used for repetition segmentation and for extracting biomechanical features. In particular, the interpolation step is useful to avoid mathematical errors like division by zero and the filtering step is essential for reliable repetition segmentation. Without applying the One Euro Filter, the raw angle signals are too unstable, which makes it difficult to detect and count repetitions accurately.

3.5.3 Repetition Segmentation

In order to provide meaningful feedback for each squat, the continuous motion data must first be divided into individual repetitions. Repetition segmentation allows the system to determine where a squat movement begins and ends, which makes it possible to analyze each repetition separately.

The segmentation is performed using the trajectory of the knee joint angle, as it provides a clear representation of the squat movement. During a squat, the knee angle decreases during the descent phase and increases again during the ascent phase. This characteristic pattern makes the knee angle a suitable signal for identifying individual repetitions.

Before detecting repetitions, the angle data is processed once more using a moving average filter:

$$\hat{a}_t = \frac{1}{w} \sum_{i=0}^{w-1} a_{t-i}$$

Where:

- a_t is the raw value of the signal at time t
- a_{t-i} represents previous signal values
- w is the number of past samples used for averaging
- $\hat{a}_t$ is the smoothed estimate of the signal at time t

This additional smoothing step reduces small fluctuations in the signal that could otherwise lead to incorrect detection of local minima or maxima. After smoothing the signal, the algorithm searches for local minima and maxima in the knee angle trajectory. A local minimum typically corresponds to the lowest point of the squat, while the surrounding maxima represent the standing position before and after the repetition.

Finally, the detected repetitions are validated using a minimum duration constraint. If the time between two detected extrema is too short, the movement is considered incomplete or invalid and is discarded.

$$T_{\text{rep}} = \frac{f_e - f_s}{FPS}$$

- T_{rep} is the minimum duration threshold
- f_s represents the start frame of the repetition
- f_e represents the end frame of the repetition

This step helps prevent small noise-induced movements from being incorrectly interpreted as full squat repetitions.

The resulting segmentation provides the boundaries for each repetition, which are then used for the feature extraction and feedback generation stages described in the following sections.

3.6 FEATURE EXTRACTION

After the squat repetitions have been segmented, several biomechanical features are extracted from each repetition. These features describe the movement characteristics of the squat and are later used to generate feedback regarding movement quality.

For each detected repetition, the algorithm analyzes the joint angle trajectories of the hip, knee, and ankle between the detected start and end frames. From these signals, several key features are computed that represent the movement depth and joint contributions during the squat.

One important feature is the minimum joint angle, which corresponds to the deepest point of the squat. The lowest knee angle θ_{min} in particular represents the bottom position of the movement and is therefore an important indicator of squat depth.

$$\theta_{\text{min}} = \min_{t \in [f_s, f_e]} \theta_t$$

- θ_t denotes the joint angle at time t

Another important feature is the range of motion (ROM) of the joint during the repetition. The ROM describes the difference between the maximum and minimum joint angle values during the movement.

$$ROM = \max_{t \in [f_s, f_e]} \theta_t - \min_{t \in [f_s, f_e]} \theta_t$$

The ROM provides information about how much the joint moves during the squat and can be used to detect shallow or incomplete repetitions.

In addition to the individual joint ROM values, the relationship between hip and knee motion can also be analyzed. For this purpose, the hip-to-knee motion ratio is calculated as

$$R_{hk} = \frac{ROM_{hip}}{ROM_{knee}}$$

Where:

- ROM_{hip} = hip joint range of motion
- ROM_{knee} = knee joint range of motion

3.7 AUTOMATIC FEEDBACK GENERATION

In this section, the extracted and processed movement data are compared with commonly recognized sports science criteria that are relevant for evaluating squat technique [Sch10; Lor+18; Esc01]. These criteria allow the system to determine whether a squat repetition is performed correctly and whether corrective feedback should be generated.

Based on these criteria, several conditions are evaluated for each detected repetition. The following aspects are considered in the feedback generation process:

- Minimal repetition duration [Sch10; KIY19]
- Insufficient knee range of motion [Esc01; KIY19; Cab+23]
- Squat performed too shallow [Esc01; KIY19]
- Excessive leg extension [Esc01]
- Excessive hip dominance [Lor+18; SP24]
- Insufficient ankle dorsiflexion [Mac+12; Lor+18; Kim+15]

These criteria are used to identify potential movement errors that may negatively affect squat performance or increase the risk of injury. For each rule, specific threshold values are defined. For example, the minimum repetition duration must be greater than one second for a movement to be considered a valid squat repetition. Previous research has shown that providing too much feedback can negatively impact performance and learning [WS90; WS02]. Therefore, in order to avoid overwhelming the athlete with excessive information, only the first detected criterion is emphasized in the generated feedback.

The thresholds used for the rule-based evaluation are summarized in Table 3.1.

Criterion	**Threshold**
Minimal repetition duration	rep duration$<$ 1.0 s
Insufficient knee range of motion	knee ROM$<$ 50.0°
Squat performed too shallow	bottom knee angle $>$ 110.0°
Excessive leg extension	knee angle $>$ 180°
Excessive hip dominance	hip/knee ratio $>$ 1.35
Insufficient ankle dorsiflexion	bottom ankle angle $>$ 100.0°

Table 3.1: Threshold values used for rule-based squat feedback generation.

The following sections provide an explanation of how and why these criteria were chosen.

Minimal repetition duration

A minimum repetition duration is used to ensure that the detected movement corresponds to an intentional squat rather than a short noise artifact or an incomplete motion. Very short or unusually fast movements can occur due to tracking errors or partial repetitions. By applying a minimum duration threshold, the system filters out such unrealistic movements and only considers repetitions that resemble a deliberate squat. In general, excessively fast movement execution can increase joint-related shear and compressive forces during the squat [Sch10].

Insufficient knee range of motion

During a proper squat, the knee joint should move through a sufficient range of motion. If the ROM is too small, the detected movement likely corresponds to a partial or shallow squat. Such movements do not represent the intended exercise and may also reduce the effectiveness of the training. For this reason, a minimum knee ROM threshold is introduced to ensure that only meaningful squat repetitions are included in the analysis.

Squat performed too shallow

Squat depth plays an important role in correct squat execution. A shallow squat occurs when the knee angle does not decrease sufficiently during the downward phase of the movement. From a training perspective, insufficient depth can reduce muscle activation and limit the overall effectiveness of the exercise. Detecting shallow squats therefore allows the system to provide feedback that encourages the user to perform the movement with greater depth.

Excessive leg extension

This rule detects repetitions in which the knees remain close to full extension throughout the movement. If the knee angle approaches or exceeds 180°, excessive stress can be placed on the knee joint. Over time, this may increase the risk of injury, particularly when performing squats with heavy loads.

Excessive hip dominance

The relationship between hip and knee motion provides insight into how a squat is performed. If the hip joint moves considerably more than the knee joint, the movement may become overly hip-dominant. This pattern is often associated with excessive forward trunk lean and limited knee flexion. Such a movement strategy can indicate suboptimal technique and may increase stress on the lower back.

Insufficient ankle dorsiflexion

Adequate ankle dorsiflexion is important for maintaining balance and proper squat technique. Limited ankle mobility can lead to compensatory patterns, such as increased forward trunk lean or premature heel lift. By analyzing the ankle joint angle, the system can identify situations where restricted ankle mobility may negatively affect squat execution.

4 EXPERIMENTS

This chapter evaluates the proposed system for automatic squat analysis using smartphone video recordings. The goal of the experiments is to assess whether the implemented pipeline can reliably extract joint angle trajectories and whether the rule-based feedback system is able to detect potential deviations in squat execution.

The evaluation focuses on two main aspects. First, the stability and usability of the pose estimation pipeline are examined by analyzing the resulting joint angle signals over time. Second, the effectiveness of the rule-based feedback system is evaluated by determining whether the extracted features allow the detection of common technique deviations during squat movements.

Based on these goals, the following research questions are defined:

- RQ1: Can smartphone-based pose estimation produce stable joint angle trajectories suitable for movement analysis?
- RQ2: Can rule-based analysis detect deviations in squat technique based on the extracted joint angle features?

The following sections describe the experimental setup, the recorded dataset, and the evaluation procedure used to answer these research questions.

4.1 EXPERIMENTAL SETUP

The video recordings used for the experiments were collected according to the data acquisition setup described in Section 3.2. The dataset consists of multiple video recordings of participants performing back squats under realistic training conditions.

Each recording contains multiple squat repetitions performed with moderate loads between 20 kg and 40 kg. The recordings were processed using the pipeline described in Chapter 3, which includes pose estimation, joint angle computation, signal filtering, repetition segmentation, and feature extraction.

The extracted joint angle trajectories were then analyzed in order to evaluate the stability of the pose estimation results and to determine whether the rule-based feedback system can detect deviations in squat technique.

4.2 DATASET

The dataset used in this study consists of recordings from 14 participants aged between 20 and 30 years, including 5 women and 9 men. All participants provided informed consent prior to the recordings and had previous athletic experience, meaning they were already familiar with performing squats. While this is beneficial for collecting high-quality movement data, it also makes the evaluation of technical error detection more challenging, as experienced athletes tend to perform the movement correctly.

To address this limitation, participants with extensive training experience were asked whether they would be willing to intentionally perform squats with specific technical errors while being recorded. These additional recordings were used to evaluate whether the proposed pipeline is capable of detecting movement errors and identifying where they occur, and reliably recognizing these deviations based on the calculated joint angles and their relationships.

The following table provides an overview of the collected data.

Dataset Property	**Value**
Number of participants	14
Age range	20–30 years
Female participants	5
Male participants	9
Total recorded videos	17
Total squat repetitions	93
Correct repetitions (OK)	72
Repetitions with deviations	21
Frame rate	30 FPS
Camera perspective	Side view (sagittal plane)

Table 4.1: Overview of the dataset used for the experiments.

4.3 EVALUATION MEASURES

The performance of the proposed system is evaluated using both quantitative and qualitative analysis. First, the stability and consistency of the extracted joint angle trajectories are assessed by visualizing the estimated angles over time and comparing them to the corresponding reference values.

In addition, several error-based evaluation metrics are computed to quantify the accuracy of the pose estimation pipeline. These include the Mean Absolute Error (MAE), Root Mean Squared Error (RMSE), and the Maximum Error (MAXE) between the estimated and reference joint angles.

Furthermore, repetition-level features extracted from the joint angle trajectories, such as minimum joint angles and joint angle ratios, are compared with the corresponding reference values in order to evaluate the reliability of the feature extraction step.

Finally, the generated feedback is evaluated by comparing the predicted feedback labels with the manually annotated reference labels. Based on this comparison, the confusion matrix components true positives (TP), true negatives (TN), false positives (FP), and false negatives (FN) are determined. Using these values, the classification metrics precision, recall, F1-score, and accuracy are calculated to assess the ability of the system to distinguish between correctly executed squats and repetitions containing technique deviations.

4.3.1 *Stability*

To qualitatively assess the stability of the pose estimation pipeline, the estimated joint angle trajectories are compared with the corresponding reference values obtained from the annotated dataset. Visual inspection of the trajectories allows the temporal consistency of the estimated signals to be evaluated and provides an intuitive understanding of how closely the estimated angles follow the reference measurements.

4.3.2 *Error Metrics*

In addition to the qualitative comparison of the joint angle trajectories, quantitative error metrics are used to evaluate the difference between the estimated joint angles and the reference values. These metrics provide a numerical assessment of how accurately the proposed system estimates the joint angles during the squat movement.

For each joint (hip, knee, and ankle), the following error metrics are computed.

Mean Absolute Error (MAE) The Mean Absolute Error measures the average absolute difference between the estimated angle $\hat{\theta}$ and the reference angle θ across all samples. It provides an intuitive measure of the typical estimation error in degrees.

$$MAE = \frac{1}{N}\sum_{i=1}^{N} |\hat{\theta}_i - \theta_i|$$

where N is the number of samples, θ_i represents the reference joint angle, and $\hat{\theta}_i$ represents the estimated joint angle.

Root Mean Squared Error (RMSE) The Root Mean Squared Error measures the square root of the mean squared difference between the estimated and reference angles. Compared to MAE, RMSE penalizes larger deviations more strongly and therefore highlights larger estimation errors.

$$RMSE = \sqrt{\frac{1}{N}\sum_{i=1}^{N}(\hat{\theta}_i - \theta_i)^2}$$

Maximum Error (MAXE) The Maximum Error represents the largest absolute deviation between the estimated and reference joint angles observed in the dataset. This metric indicates the worst-case estimation error.

$$MAXE = \max_{i=1,\dots,N} |\hat{\theta}_i - \theta_i|$$

The error metrics are computed separately for the hip, knee, and ankle joint angles in order to analyze the estimation accuracy for each joint individually. Together, these metrics provide a quantitative evaluation of the accuracy of the estimated joint angle trajectories.

4.3.3 *Feedback Classification Metrics*

Since all participants had prior squat experience, most recorded repetitions were performed with correct technique. As a result, only a limited number of repetitions containing technical deviations were available for evaluating the feedback system.

To partially address this limitation, some participants were asked to intentionally perform squat repetitions with specific technical errors while being recorded. These additional recordings were used to assess whether the proposed rule-based approach is capable of identifying deviations in squat technique.

Due to the limited number of repetitions containing technical errors, the resulting classification metrics should be interpreted as indicative results rather than statistically robust performance estimates. Nevertheless, they provide insight into whether the system is generally capable of distinguishing between correct and incorrect squat executions.

For this evaluation, repetitions labeled with the feedback code `OK` are treated as technically correct executions, while all other feedback codes are interpreted as repetitions containing technical deviations. Based on this binary formulation, the predicted feedback labels are compared with the reference annotations.

From this comparison, the confusion matrix components true positives (TP), true negatives (TN), false positives (FP), and false negatives (FN) are determined. Using these values, the commonly used classification metrics precision, recall, accuracy and F1-score are calculated in order to evaluate the performance of the feedback detection system.

	Predicted Positive	**Predicted Negative**
Actual Positive	True Positive (TP)	False Negative (FN)
Actual Negative	False Positive (FP)	True Negative (TN)

Table 4.2: Confusion matrix used for evaluating the feedback classification.

Based on the values obtained from the confusion matrix, the evaluation metrics precision, recall, accuracy and F1-score are calculated as follows:

$$Precision = \frac{TP}{TP + FP}$$

$$Recall = \frac{TP}{TP + FN}$$

$$Accuracy = \frac{TP + TN}{TP + TN + FP + FN}$$

$$F1 = 2 \cdot \frac{Precision \cdot Recall}{Precision + Recall}$$

4.4 RESULTS

This section presents the experimental results of the proposed system. The evaluation focuses on two main aspects. First, the accuracy and stability of the estimated joint angle trajectories are analyzed. Second, the performance of the rule-based feedback system in detecting technical deviations during squat execution is evaluated.

4.4.1 *Joint Angle Trajectories*

Figure 4.1 presents a comparison between the estimated joint angles and the corresponding reference measurements for the hip, knee, and ankle joints during a squat sequence performed by an experienced participant. The x-axis represents time in seconds, while the y-axis indicates the joint angles in degrees.

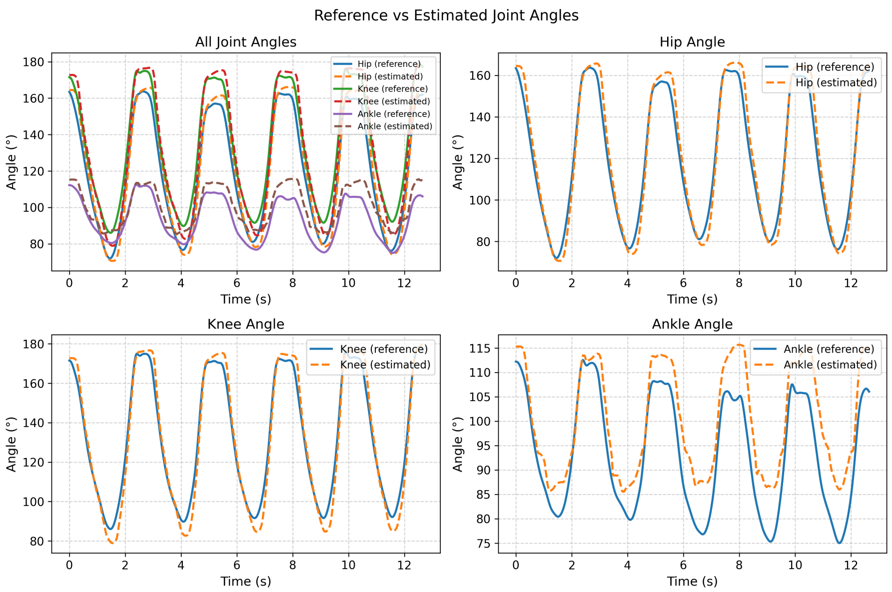

Figure 4.1: Comparison of reference and estimated joint angles for hip, knee, and ankle during a squat repetition with correct technique.

The estimated trajectories closely follow the reference measurements for all three joints throughout the squat movement. The general shape and timing of the movement phases are captured well by the estimated signals, including the descent, bottom position, and ascent of the squat.

As expected from squat biomechanics, the knee joint exhibits the largest range of motion during the movement. The hip joint shows a slightly smaller but still pronounced range of motion, while the ankle joint changes more gradually over time.

Overall, the estimated signals appear smooth and consistent, indicating that the pose estimation pipeline provides stable joint angle estimates suitable for further analysis. Small deviations between the reference and estimated signals can be observed at some peak positions, which is expected due to minor landmark detection inaccuracies.

4.4.2 *Joint Angle Error Metrics*

To quantitatively evaluate the accuracy of the estimated joint angles, the mean absolute error (MAE), root mean squared error (RMSE), and maximum error (MAXE) were calculated for the hip, knee, and ankle joints across all participants.

Figure 4.2 presents the aggregated error metrics for the three joints. Overall, the results show relatively small average prediction errors, indicating that the proposed pose estimation pipeline is able to estimate joint angles with sufficient accuracy for movement analysis.

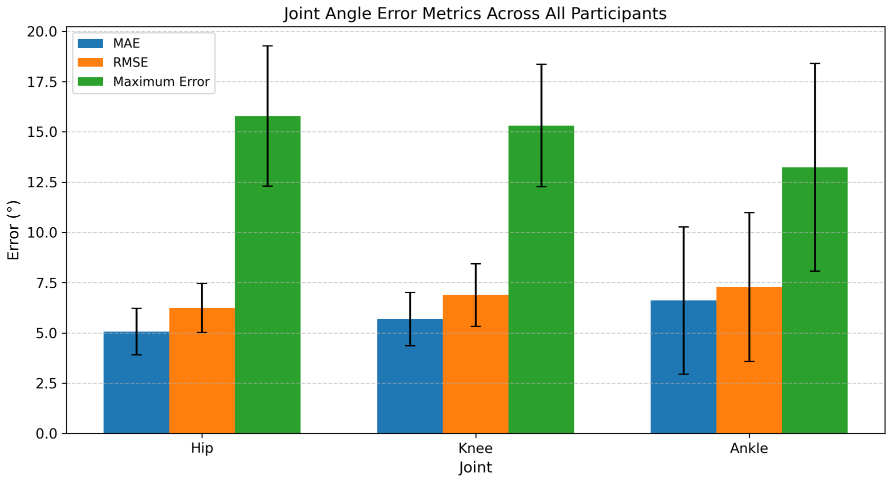

Figure 4.2: Joint Metrics

The hip and knee joints show similar average error values, with MAE values around five to six degrees. The ankle joint exhibits comparable errors but slightly larger variability, which can be observed in the larger standard deviation.

Overall, the magnitude of the errors is small compared to the typical joint angle ranges observed during squat movements. This suggests that the estimated joint angles are sufficiently precise to support the subsequent repetition feature extraction.

4.4.3 *Repetition Feature Extraction*

To analyze individual squat repetitions, several repetition-level features were extracted from the joint angle signals. These features include the minimum knee angle, minimum ankle angle, minimum hip angle, and the hip-to-knee ratio.

Figure 4.3 shows a comparison between the reference features and the corresponding estimated values for multiple repetitions.

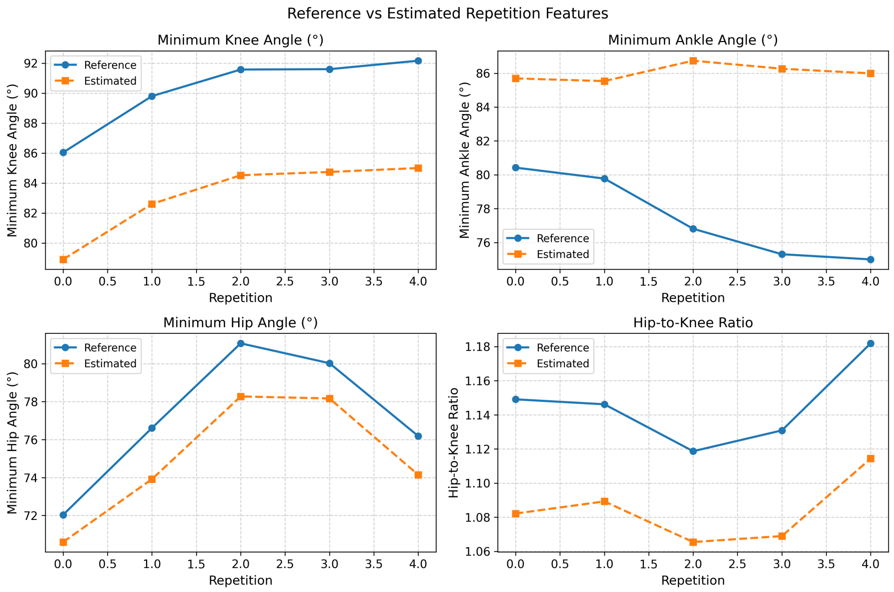

Figure 4.3: Comparison between reference and estimated repetition features.

The results show that the estimated repetition features generally follow similar trends to the reference measurements across the recorded repetitions. Although noticeable deviations between the estimated and reference values are present, particularly for some joint angles, the overall progression of the features across repetitions remains comparable.

In particular, the minimum joint angles extracted from the estimated trajectories still reflect changes in squat depth and movement behavior between repetitions. Although the absolute values sometimes differ from the reference measurements, the relative changes over repetitions are largely preserved.

This suggests that the proposed pipeline is capable of extracting biomechanically meaningful features from the estimated joint angles, although the precision of the estimated values is limited in some cases. The observed deviations highlight that the extracted features should be interpreted with caution, especially when relying on absolute angle values rather than relative trends.

4.4.4 Feedback Classification Results

The final step of the pipeline assess whether the system can correctly identify technical deviations during squat execution. The predicted feedback labels were compared with the manually annotated reference labels for each repetition.

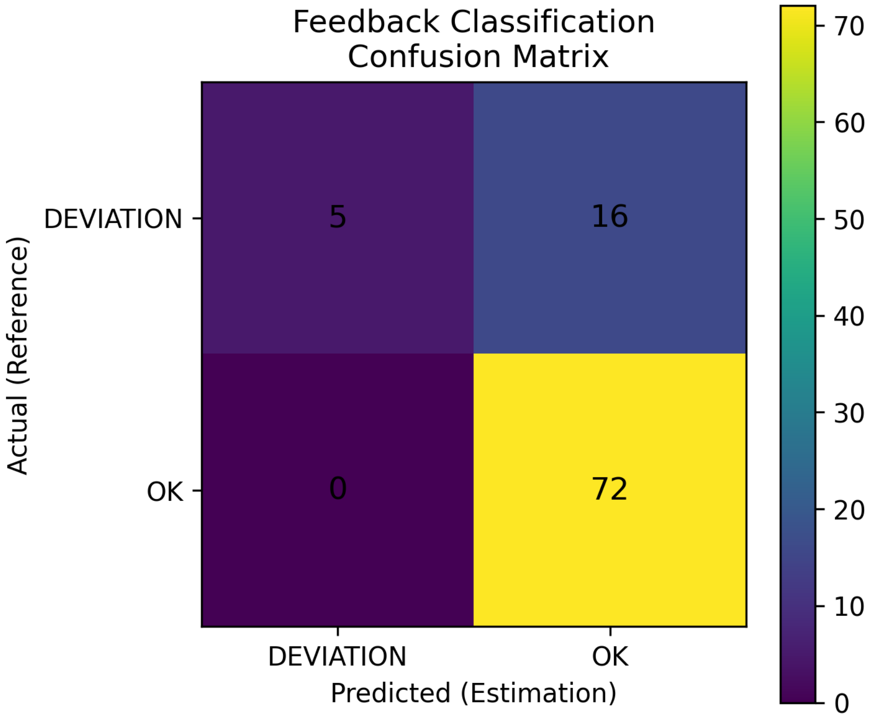

Figure 4.4: Confusion matrix for the feedback classification.

Figure 4.4 shows the resulting confusion matrix. The system correctly classified 72 repetitions as technically correct and 5 repetitions as containing deviations. In addition, 16 repetitions containing deviations were incorrectly classified as correct, while no correct repetitions were misclassified as deviations.

Based on the confusion matrix, precision, recall, accuracy, and F1-score were calculated.

The corresponding evaluation metrics are shown in Figure 4.5. The system achieves a precision of 1.0, a recall of 0.238, an accuracy of 0.828, and an F1-score of 0.385. However, the relatively low recall indicates that the system fails to detect a considerable number of incorrect squat executions. This suggests that the rule-based thresholds may be too conservative.

This behavior can be explained by the limited number of deviation examples in the dataset as well as the conservative design of the rule-based feedback system. Since the feedback rules are based on predefined thresholds, subtle deviations that do not exceed these limits may remain undetected. Additionally, this effect is influenced by the

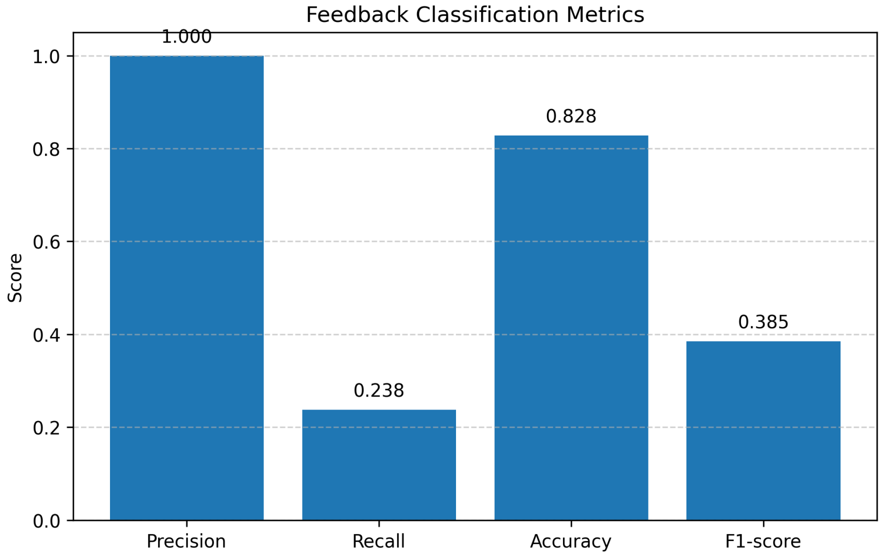

Figure 4.5: Feedback classification metrics.

imbalance of the dataset, as the majority of recorded repetitions were performed with correct technique.

4.4.5 *Threshold Optimization and Error Analysis*

To evaluate whether the performance of the rule-based feedback system could be improved, the threshold values of all feedback criteria were re-evaluated across the entire dataset. A grid search was performed over the defined threshold ranges to identify a parameter configuration that maximizes the F1-score as shown in Figure 4.6.

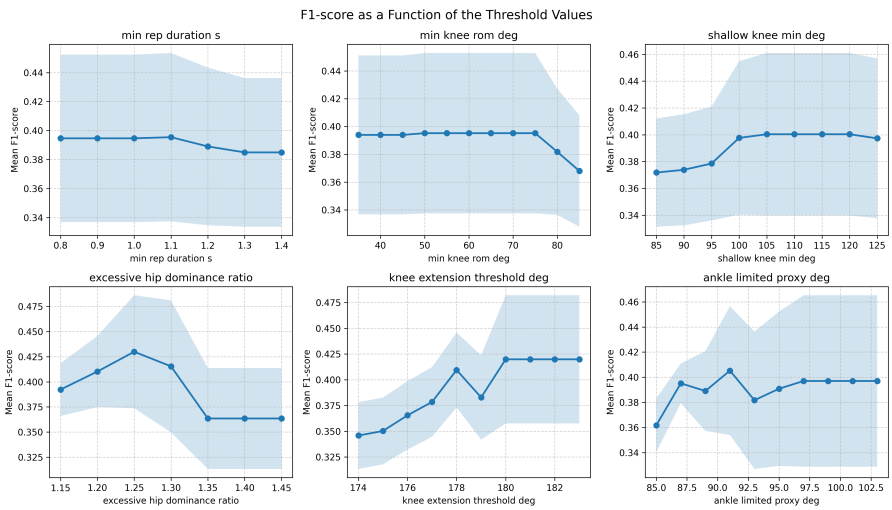

Figure 4.6: F1-Score vs Threshold Values

The optimized parameter configuration, presented in Table 4.3, resulted in a modest improvement in classification performance. As

Criterion	Threshold
Minimal repetition duration	rep duration$<$ 1.1 s
Insufficient knee range of motion	knee ROM$<$ 50.0°
Squat performed too shallow	bottom knee angle $>$ 105.0°
Excessive leg extension	knee angle $>$ 180°
Excessive hip dominance	hip/knee ratio $>$ 1.25
Insufficient ankle dorsiflexion	bottom ankle angle $>$ 91.0°

Table 4.3: Threshold values after optimization.

shown in Figure 4.7, the F1-score increased from 0.385 to 0.526. While recall improved to 0.588, precision decreased to 0.476, and the overall accuracy slightly decreased to 0.806. This indicates that the optimized thresholds allow the system to detect a larger number of deviations, but at the cost of an increased number of false positive classifications.

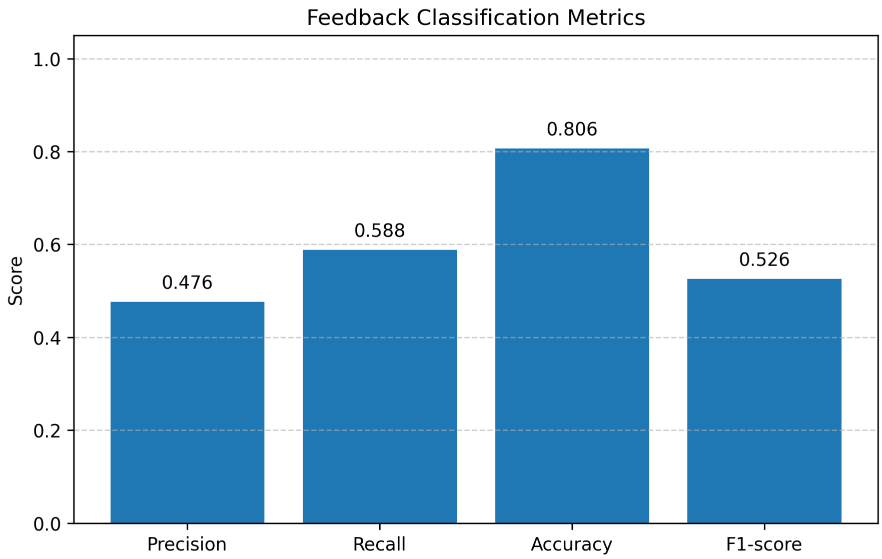

Figure 4.7: Feedback classification metrics after optimization.

A closer inspection of the misclassified repetitions revealed that a considerable portion of the remaining errors correspond to a specific technical mistake in which the athlete lifts the heel off the ground during the squat as shown in Figure 5.1, Heel Error Not Detected. This movement error cannot be reliably detected using only the three joint angles (hip, knee, and ankle) extracted by the pose estimation pipeline. Since the heel landmark itself is not explicitly considered in the current feature set, the pipeline lacks the necessary information to identify this deviation.

To further investigate this limitation, all repetitions that contain this specific error were excluded from the evaluation. The resulting confusion matrix is shown in Figure 4.8. After removing these samples, classification performance improves substantially, achieving perfect scores for precision, recall, accuracy, and F1-score.

These results suggest that the rule-based feedback system is capa-

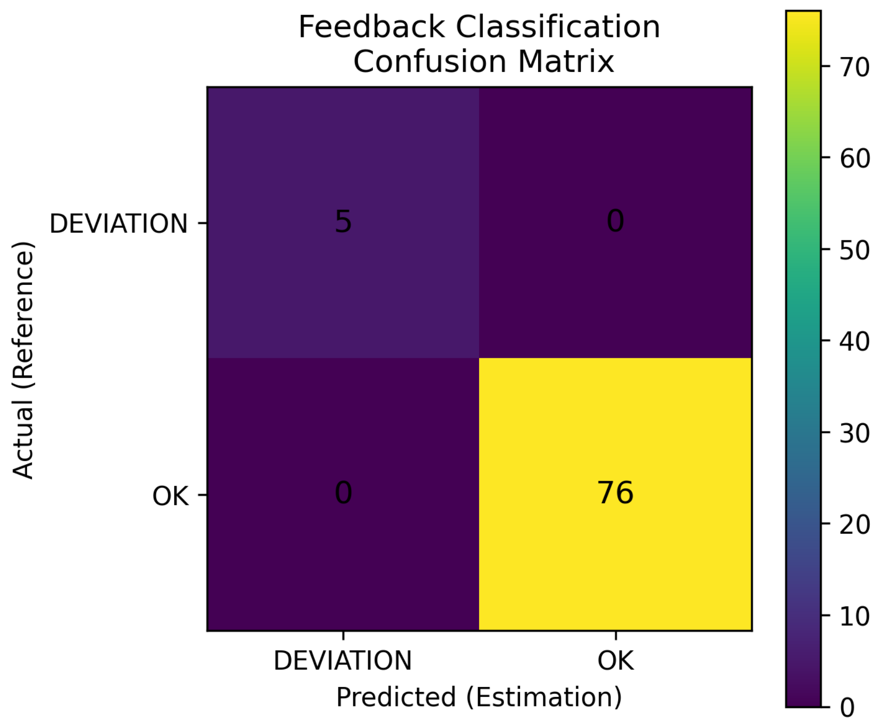

Figure 4.8: Confusion matrix for the feedback classification after excluding the 12 repetitions containing the specific error.

ble of reliably detecting the remaining types of movement deviations present in the dataset. At the same time, they highlight a limitation of the current feature representation. Detecting heel lift errors would likely require the inclusion of additional pose landmarks, such as heel or foot position, or the use of a more detailed biomechanical model of the foot–ground interaction.

4.4.6 *Example Feedback Visualization*

In addition to the quantitative evaluation, visual examples of the generated feedback were inspected. Figure 4.9 shows an example frame from the annotated output video produced by the system. The pose estimation model detects the body landmarks and overlays the estimated skeleton together with the calculated joint angles and the corresponding feedback message on the top.

The visualizations demonstrate how the system provides feedback during squat movement. The estimated joint angles are displayed together with the detected pose skeleton, allowing movement execution to be analyzed directly from the video output.

Figure 4.9: Example frame from the annotated output video showing the pose estimation and generated feedback.

5 CONCLUSION

This thesis investigated whether smartphone video data combined with pose estimation can be used to automatically analyze squat executions and generate technique-related feedback. A pipeline was developed that estimates body landmarks using MediaPipe Pose, calculates joint angles for the hip, knee, and ankle, segments individual squat repetitions, and applies rule-based criteria to identify potential technique deviations.

The experimental evaluation shows that the proposed approach is capable of producing stable joint angle trajectories from side-view video recordings. The estimated angles closely follow the reference measurements, and the overall prediction errors remain relatively small. These results indicate that markerless pose estimation can provide sufficiently accurate joint angle estimates for movement analysis using consumer-grade devices.

In addition, repetition-level features extracted from the estimated joint angles were successfully used to analyze squat execution. The rule-based feedback system was able to distinguish between correctly executed repetitions and repetitions that contain technique deviations. Although the system performs well in identifying stable and correct squat movements, some deviations remain difficult to detect reliably.

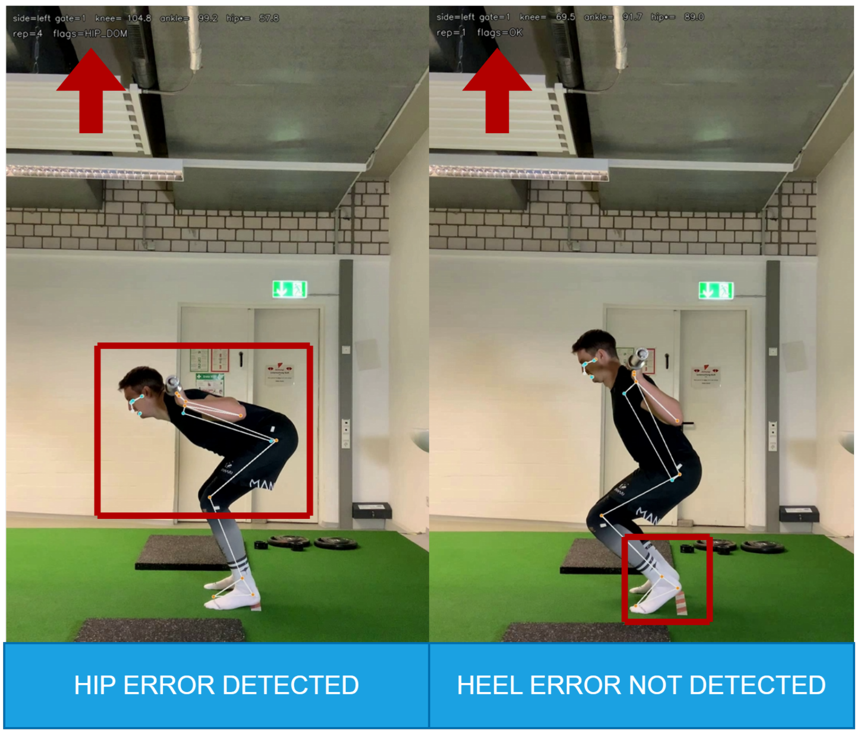

Figure 5.1: Example of a detected and not detected deviation.

5.1 LIMITATIONS

Several limitations of the proposed approach must be considered.

First, the analysis is based on two-dimensional pose estimation, which means that movements outside the sagittal plane cannot be accurately captured. Therefore, the system may fail to detect certain deviations in the frontal or transverse plane.

Second, the evaluation focuses on a lateral view of the squat movement, although this perspective is suitable for analyzing hip, knee, and ankle movement, heel lift during a squat cannot be reliably detected using joint angles alone, as shown in Figure 5.1.

Furthermore, occlusions caused by weight plates or variations in clothing may affect the accuracy of the detected landmarks. However, in this study, these factors had only a minor impact due to the relatively controlled recording conditions.

Finally, the dataset used in this study is relatively small and contains only a limited number of repetitions with technical variations. This makes it difficult to draw statistically robust conclusions about the performance of the feedback system.

5.2 FUTURE WORK

Future work could address several of the limitations identified in this study. Collecting a larger and more diverse dataset would allow a more comprehensive evaluation of the system, particularly to detect a wider range of movement errors.

In addition, incorporating additional camera perspectives or using three-dimensional pose estimation could improve the detection of more complex movement deviations. Tracking additional landmarks, such as the heel position, may also help identify specific technique errors that cannot be detected with the current setup.

Another promising direction is the integration of machine learning approaches for feedback generation. Instead of relying solely on predefined rule-based thresholds, data-driven models could learn more complex relationships between joint angles and movement quality.

5.3 FINAL WORDS

Overall, the results of this thesis indicate that smartphone-based pose estimation provides a promising foundation for automatic movement analysis and feedback generation in strength training exercises such as the squat. However, the potential of this approach is not limited to squat analysis alone. In principle, the proposed method could be extended to analyze a wide range of strength training exercises beyond the squat, allowing automated assessment of movement technique in different types of workouts. By providing accessible feedback on movement execution, such systems can also contribute to reducing the risk of long-term injuries caused by repeatedly performing exercises with improper technique.

These findings highlight the potential of accessible computer vision tools to help athletes and coaches analyze movement execution without requiring specialized motion capture equipment. With further improvements in pose estimation accuracy and real-time processing, such systems could become valuable tools for everyday training environments.

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